

Max Young (Le Yang)

Senior Software Engineer /
Autonomous Driving
SOTA/FOTA & Fleet Platforms

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Profile

Senior software engineer with 13+ years of experience building production systems, including 4+ years in autonomous driving. Leads vehicle-platform work from product definition and architecture through implementation and fleet operations. Reduced average SOTA deployment time by 67% while expanding capacity 25× from 200 to 5,000 vehicles, and led an autonomous-vehicle FOTA platform to M3/MVP. Strong in Python, React and Linux, with practical C++ experience; open to relocation across Germany.

Professional Experience

2022–Present **Software Engineer, Rino.ai**, Beijing, China, Autonomous Driving

- Led architecture, product definition and technical direction for a three-person vehicle-platform team, turning separate engineering tools into reusable delivery infrastructure for R&D, factory and fleet operations.
- **SOTA Platform:** Reduced average software deployment time from approximately 30 to 10 minutes (67%) while increasing maximum capacity from 200 to 5,000 vehicles (25×).
- **FOTA Platform:** Led design and implementation through M3/MVP, covering distributed execution across TBOX/ADCU nodes, upgrade orchestration, failure recovery, power-loss safety, vehicle clients, backend deployment and web-based management.
- Unified delivery workflows for software, maps, vehicle configuration and calibration, including vehicle/series targeting, execution verification, service-health monitoring, interrupted-transfer recovery and version/state consistency.
- **Large-map Distribution:** Led architecture trade-offs for hundreds-of-gigabytes datasets under limited disk space and unreliable networks, evaluating rsync, incremental delivery and OverlayFS for storage efficiency, atomic switching, rollback, progress and ETA accuracy.
- **Reliability & Observability:** Improved fleet-update transparency by analysing rsync transfer-state semantics, interrupted-transfer behaviour and ETA accuracy, and by exploring lightweight health monitoring for vehicle-hosted services across several thousand vehicles.
- **Vehicle Calibration Platform:** Created an end-to-end Python, Shell, C++ and OpenCV workflow used on hundreds of vehicles; increased factory throughput from 2–3 to 10+ vehicles per day and cut field calibration from 0.5–1 day to under 30 minutes.
- **Fleet Operations Platform:** Built a Flask/React/Ant Design system that standardised incident management and route delivery across R&D and operations; helped reduce the operational incident rate from 13 to 7 per 10,000 km within six months and supported delivery of dozens of new routes.
- **Engineering Infrastructure:** Evaluated GitLab's PyPI Registry for reusable internal Python packages and improved deployment automation using systemd timers, controlled long-running rsync processes, distributed FRP and cloud-hosted backend/web services.
- **Productisation:** Defined the product positioning, web information architecture and maintainable documentation approach for a unified vehicle-management platform, turning project-specific tooling into reusable team and product capabilities.

Professional Experience (continued)

- 2017–2022 **Software Engineer**, *Dedao*, Beijing, China, Online Education & E-commerce
- Designed backend systems for inventory, orders and distribution; automated order ingestion and settlement across JD.com, Taobao and Youzan using Python/Django and Go/Gin.
 - Built financial recognition, settlement and analytics services in Flask, replacing manual office workflows and improving finance-team efficiency.
 - Introduced Celery/RabbitMQ and Faust/Kafka for asynchronous processing, built a GraphQL gateway for service integration, and championed TDD and contract testing to reduce defects and integration cost.
- 2014–2017 **Software Engineer**, *Kaoshixing*, Beijing, China, B2B SaaS
- Led architecture, full-stack development, DevOps and engineering coordination for an online examination platform, taking it from initial build to production scale serving 10,000+ enterprise customers. *Python, Flask, JavaScript, Linux*
- 2013–2014 **Map Quality Engineer**, *AutoNavi (Alibaba Group)*, Beijing, China, Mapping Services
- Analysed geospatial POI and road-network data, established map-asset quality-control standards, and advised data-collection planning and quality measurement.
- 2006–2013 **Geomatics Engineer**, *Beijing Engineering Corporation*, Beijing, China, Hydropower Engineering
- Delivered geospatial surveying and technical design for large-scale hydropower infrastructure projects in China and overseas.

Technical Skills

- Domain Autonomous-driving operations, SOTA/FOTA lifecycle and release orchestration, fleet platforms, software/map/configuration/calibration delivery, observability and recovery
- Backend & Data Python (Flask, Django), REST APIs, Celery, RabbitMQ, Kafka/Faust, MQTT/EMQX, MySQL, MongoDB
- Frontend React, Next.js, Ant Design, HTML/CSS; end-to-end product and workflow design
- Systems Linux, Shell, Docker, Git, CI/CD, systemd, rsync, OverlayFS, FRP; deployment, monitoring, failure handling and operational support
- Additional Go (Gin), practical C++ and OpenCV, TDD, contract testing, agile delivery

Education

- 2002–2006 **Bachelor's Degree in Geomatics Engineering**, *Hohai University*, Nanjing, China

Languages & Additional Information

- Languages English (IELTS Academic 6.5); Japanese (JLPT N2); Mandarin Chinese (native)
- Mobility Open to relocation across Germany and to working in international teams
- Interests Distance running; certified third-level amateur marathon athlete in Beijing